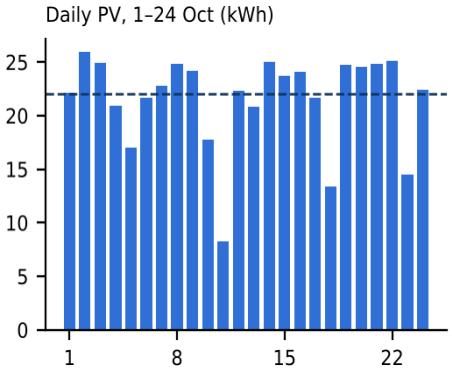
Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

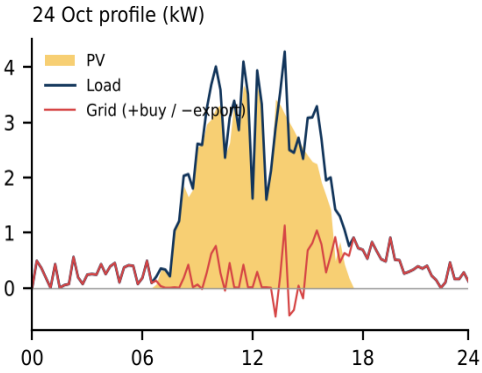
Health centre, Chiang Mai

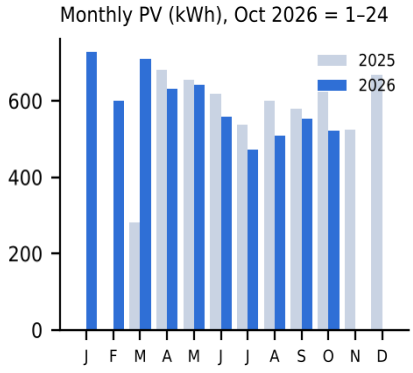
Upgrade opportunity

Load is double the PV – add panels

5.12 kWp	517 kWh	4.21	1,105 kWh	589 kWh	1 kWh
PV · 5 kW inverter	PV, 1–24 Oct	kWh/kWp/day	Consumption	Bought from grid	Exported







What we found

• 5.12 kWp with optimizers made 21.6 kWh a day (4.2 kWh/kWp), close to what we expect. All 8 optimizers are working evenly.

• The health centre used 1,105 kWh in 24 days but PV covered only 47%. It bought 589 kWh and exported almost nothing (1.2 kWh).

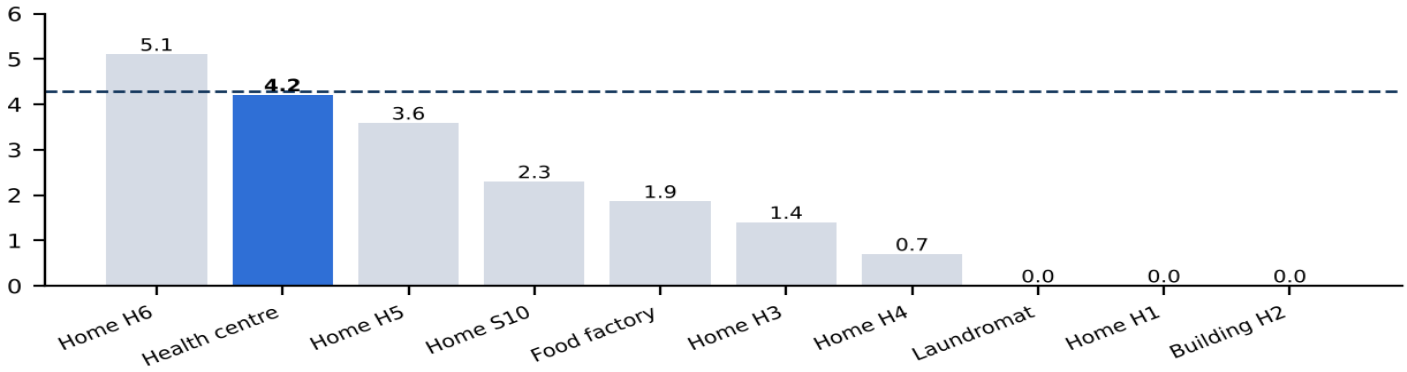
• At midday PV follows the load, with the meter close to 0 kW, so the zero-export limit is holding output back slightly on low-load days.

What to do

1. Propose about 5 kWp more PV with a second inverter (the existing 5 kW three-phase inverter is fully used). That would cover about 600 kWh a month more and save around 2,700 THB a month.

2. Keep zero export on. If the panels are added, check the daytime load first, as weekend load is lower.

How this plant compares (kWh/kWp per day, 1–24 Oct)



Rank 2 of 10 Huawei plants. Dashed line = expected 4.3 kWh/kWp per day.